

Fig. 3: Equivalent electrical circuit during fluorescent pre-heat cycle

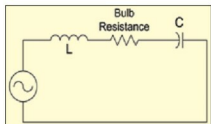


Fig. 4: Equivalent electrical circuit after strike

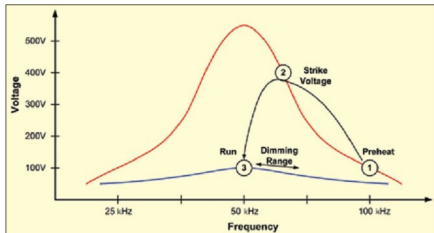


Fig. 5: Electronic fluorescent ballast operation

heated, less stress placed on the filaments results in longer bulb life.

Electronic ballasts

Fig. 2 shows the circuit that is commonly used to implement an electronic ballast, which offers better efficiency than inductor ballasts.

A resonant circuit using an inductor and capacitor is placed in series with the lamp. The L-C circuit values are chosen such as to provide resonance at the switching frequency, when the lamp is operating. Typically, electronic ballasts use switching frequencies in the range of 20-50 kHz. A second capacitor placed across the lamp effectively connects the two

filaments in series at frequencies higher than the resonant frequency. Typically, the starting frequency is 100 kHz.

During the filament heating cycle (high-frequency), C_f conducts and the circuit looks as shown in Fig. 3. After certain time duration of filament heating, the frequency is reduced to start the lamp. C_f becomes an open circuit at the lower frequency. Before the lamp starts, the resistance of the bulb is high and the voltage rapidly rises to the strike voltage, due to the resonance of the inductor-capacitor circuit. The strike voltage

is typically 600V AC or higher.

After the arc is struck, the resistance of the bulb falls and the equivalent circuit looks as shown in Fig. 4. The voltage across the bulb also drops dramatically and is usually less than 100V AC.

Fig. 5 shows the graphical representation of the operating stages of an electronic fluorescent ballast.

Generating a precise variable frequency from a digital PWM module

A typical pulse-width-modulation (PWM) peripheral on an MCU can easily be used to generate variable-frequency oscillation (VFO) required for

ballast applications. The PWM module is well suited for generating variable pulse widths at a fixed frequency, but may not have the frequency precision required for the ballast application. This is especially true if dimming is required. The dimming of a fluorescent bulb can be accomplished by increasing the frequency away from the resonant frequency of the drive circuit. Fortunately, there is an easy way to solve the problem using software.

Fig. 6 shows a typical digital PWM peripheral found on an MCU. There is a digital timebase and associated period register that sets the PWM signal period. A third register is used to set the duty cycle of the PWM.

To make the things easy, let us assume that the timebase is clocked from a 10MHz source. Further, assume that the centre frequency required for the ballast circuit is 50 kHz. A value of 200 would be loaded into the period register to get a 50kHz output signal. For good dimming adjustments, you need to be able to vary the frequency in 0.1 per cent (50Hz) steps. If the period register changes to '199', an output frequency of 50,251 Hz results. So we need a way to increase the frequency resolution by a factor of '5'.

The frequency resolution can be increased by forcing the PWM module to switch between two adjacent periods over time. At the end of each PWM period, a value is added to an accumulator register. The value added to the register represents the fractional portion of the required period. If the register overflows as a result of the addition, the higher of the period values is written to the PWM module in order to produce a lower frequency. If the register does not overflow, the lower period value is used.

Assuming that the PWM module has an 8-bit period register and the accumulator register is also 8-bit, you effectively have a 16-bit register for setting the PWM period. Stated another way, you have added eight fractional bits to the frequency-adjustment resolution.